Detailed Frontend Roadmap for AI-Powered IDE Project

1. HTML5

Why We Use It: HTML is used to create the structure of your frontend application like buttons, forms, pages, editor containers, chat sections, dashboards, and navigation menus.

Topics to Learn Deeply:
• Semantic tags (header, footer, section, article)
• Forms and validations
• Tables and lists
• Audio/video embedding
• Accessibility
• Local storage and session storage
• SEO basics

Best Source: https://developer.mozilla.org/en-US/docs/Web/HTML

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2. CSS3

Why We Use It: CSS is used for designing the frontend UI, responsiveness, layouts, animations, themes, and making the IDE visually modern.

Topics to Learn Deeply:
• Flexbox
• CSS Grid
• Responsive design
• Media queries
• Transitions and animations
• Dark/light themes
• Positioning

Best Source: https://developer.mozilla.org/en-US/docs/Web/CSS

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3. JavaScript

Why We Use It: JavaScript adds functionality and interactivity to the IDE such as file opening, chat interactions, button actions, API calls, and dynamic rendering.

Topics to Learn Deeply:
• DOM manipulation
• Closures and scope
• Async/Await
• Promises
• Fetch API

• ES6+ features
• Event handling
• Debouncing and throttling

Best Source: https://javascript.info/

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4. TypeScript

Why We Use It: TypeScript helps build large-scale applications with fewer bugs using strict typing and better code maintainability.

Topics to Learn Deeply:
• Types and interfaces
• Generics
• Enums
• Utility types
• React typing
• API typing

Best Source: https://www.typescriptlang.org/docs/

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5. React.js

Why We Use It: React is the core frontend framework used to build reusable UI components like editor windows, chat panels, dashboards, and terminal screens.

Topics to Learn Deeply:
• Components
• Props and state
• Hooks
• Context API
• Routing
• Lifecycle methods
• Performance optimization

Best Source: https://react.dev/learn

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6. Tailwind CSS

Why We Use It: Tailwind helps create fast modern UI designs with utility classes and responsive layouts.

Topics to Learn Deeply:
• Utility classes
• Responsive utilities
• Dark mode
• Custom themes
• Component styling

Best Source: https://tailwindcss.com/docs

7. Redux Toolkit

Why We Use It: Redux Toolkit manages global application state such as authentication data, opened files, AI responses, and user settings.

Topics to Learn Deeply:
• Store setup
• Slices
• Reducers
• Async thunk
• Middleware

Best Source: https://redux-toolkit.js.org/

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8. API Integration

Why We Use It: API integration connects frontend with backend services and AI models.

Topics to Learn Deeply:
• REST APIs
• Axios
• Fetch
• Authentication tokens
• Error handling
• Loading states

Best Source: https://axios-http.com/docs/intro

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9. Socket.IO

Why We Use It: Socket.IO enables real-time communication for AI streaming, live collaboration, notifications, and chat systems.

Topics to Learn Deeply:
• WebSockets
• Live updates
• Streaming
• Event-based communication

Best Source: https://socket.io/docs/v4/

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10. Monaco Editor

Why We Use It: Monaco Editor is used to build the code editing experience similar to VS Code.

Topics to Learn Deeply:
• Syntax highlighting
• Autocomplete
• Themes
• Multi-language support
• File tabs

Best Source: https://microsoft.github.io/monaco-editor/

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11. xterm.js

Why We Use It: xterm.js creates browser-based terminal UI for command execution and shell access.

Topics to Learn Deeply:
• Terminal rendering
• Shell integration
• Command streaming

Best Source: https://xtermjs.org/

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12. Framer Motion

Why We Use It: Framer Motion is used for smooth animations, transitions, sidebar effects, and interactive frontend experiences.

Topics to Learn Deeply:
• Page transitions
• Animations
• Drag and drop
• Loading animations

Best Source: https://www.framer.com/motion/

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13. Performance Optimization

Why We Use It: Performance optimization ensures the IDE runs smoothly even with multiple files and AI responses.

Topics to Learn Deeply:
• Lazy loading
• Code splitting
• Memoization
• Virtualization

Best Source: https://react.dev/reference/react/memo

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14. Deployment

Why We Use It: Deployment makes your frontend accessible online for users.

Topics to Learn Deeply:
• Vercel deployment
• Environment variables
• Build optimization
• Production setup

Best Source: https://vercel.com/docs